# SMART FIRE DETECTION & CONTROL BOT

# TEAM MEMBERS

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# PRESENTATION OVERVIEW

# 1. INTRODUCTION

The Smart Fire Detection and Control Bot is an automated system designed to detect and extinguish fire with minimal human intervention. It uses flame sensors to identify fire and a microcontroller to process the input and control system operations. The robot is capable of moving towards the fire source using motor-driven wheels and responding quickly to reduce fire spread. A water pump mechanism is used to spray water and suppress the fire effectively.

The system is designed to be simple, cost-effective, and suitable for small-scale applications such as homes and laboratories. It combines detection, mobility, and fire suppression into a single integrated system. The project demonstrates the practical use of embedded systems and robotics in improving fire safety and reducing risks to human life.

# 2. PROBLEM STATMENT

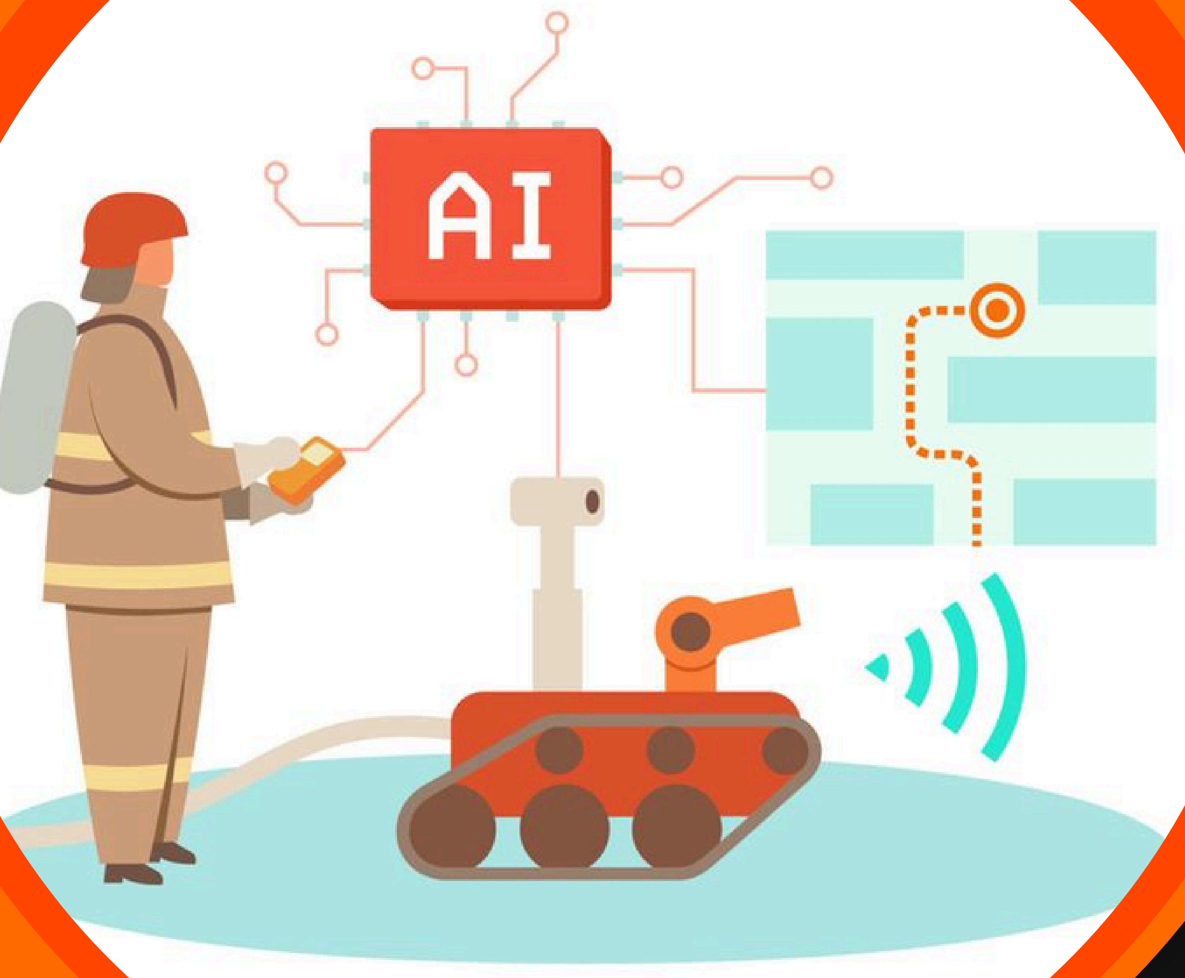
Fire accidents can cause severe damage to life and property, especially when not detected at an early stage. Most existing systems rely only on alerts and require human intervention, which leads to delayed response and increased risk.

**Delayed Detection**

**Slow Response**

**No Automatic Control**

**Risk to Human Life**




# 3. OBJECTIVES

The main objective of this project is to design and develop an automated fire detection and control system that ensures quick response and improved safety. The system aims to continuously monitor the environment, detect fire at an early stage, and take immediate action without human intervention. It also focuses on reducing risks to human life and minimizing damage caused by fire accidents.

- Fire Detection
- Automatic Movement
- Fire Suppression
- Quick Response

# 4. PROPOSED SYSTEM

The proposed system is a Smart Fire Detection and Control Bot that integrates sensing, control, and actuation into a single automated system. It uses flame sensors to detect fire and a microcontroller to process the signals and control the robot's movement. Based on the detected fire location, the robot navigates towards the source and activates a water pump to extinguish the fire efficiently.

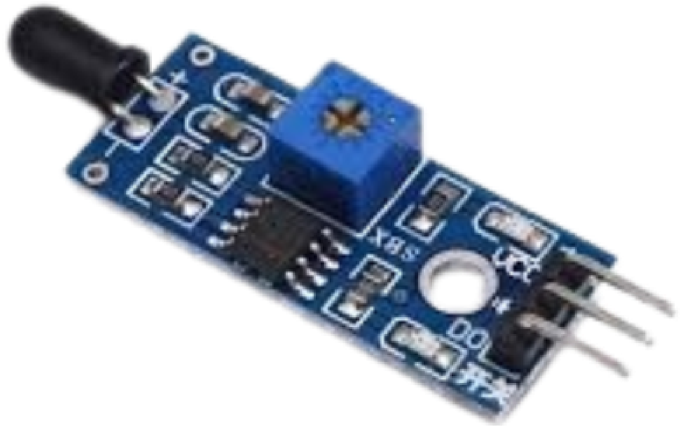
Sensor + Controller

Movement Mechanism

Extinguishing Unit

# 5. COMPONENTS


Arduino UNO

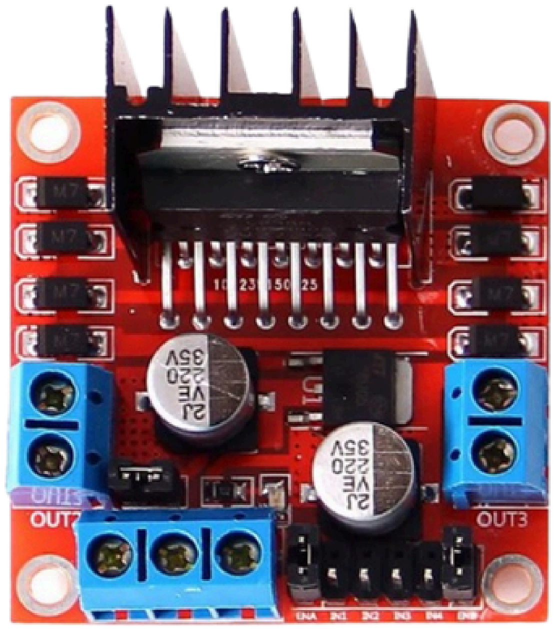


IR Flame Sensor


Relay Module

Double Layer Smart Car Chassis with motors

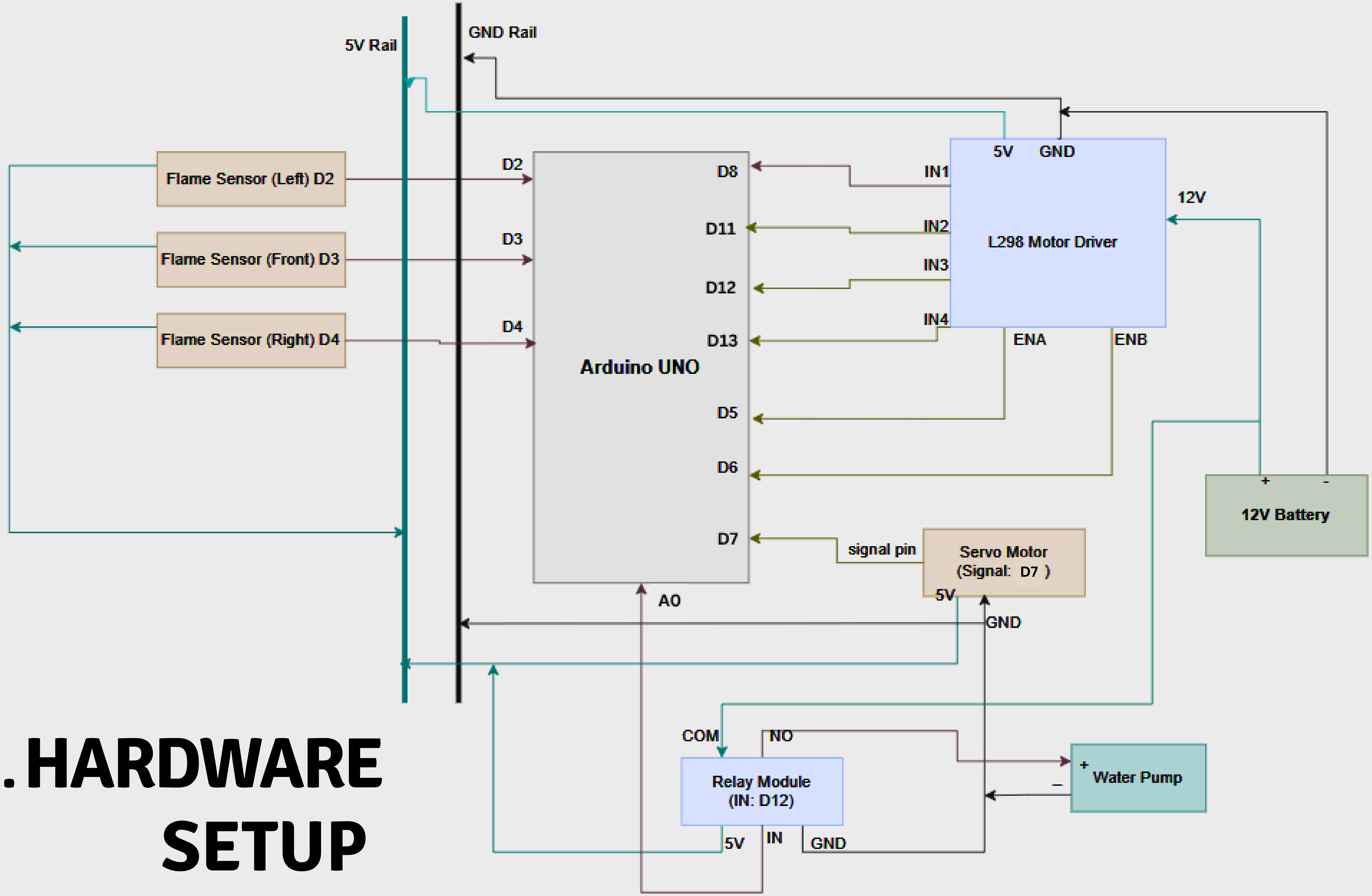

Servo S90G




L298N Motor Driver

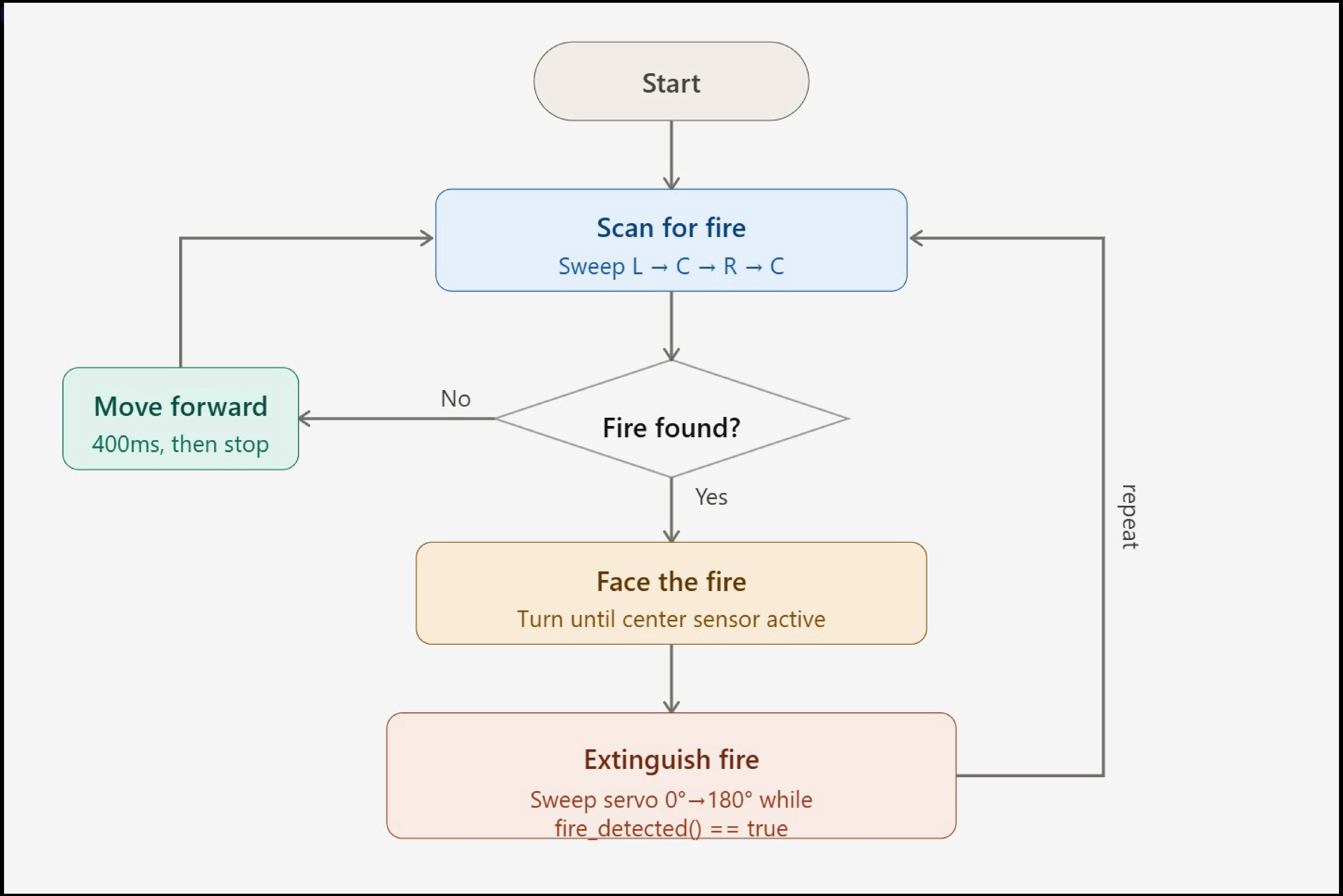
5V DC Mini Water Pump Motor

12V Battery



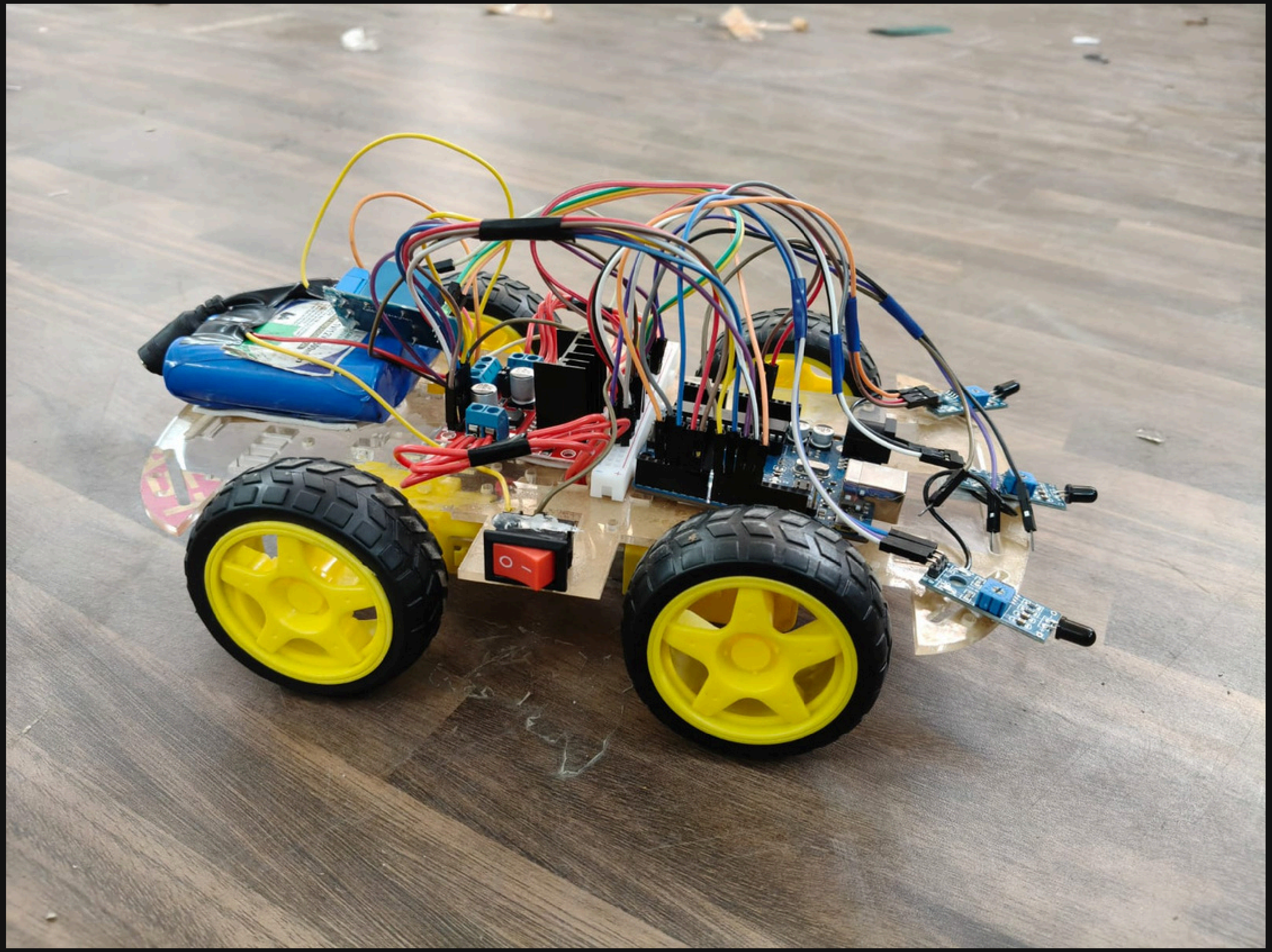

# 6. HARDWARE SETUP

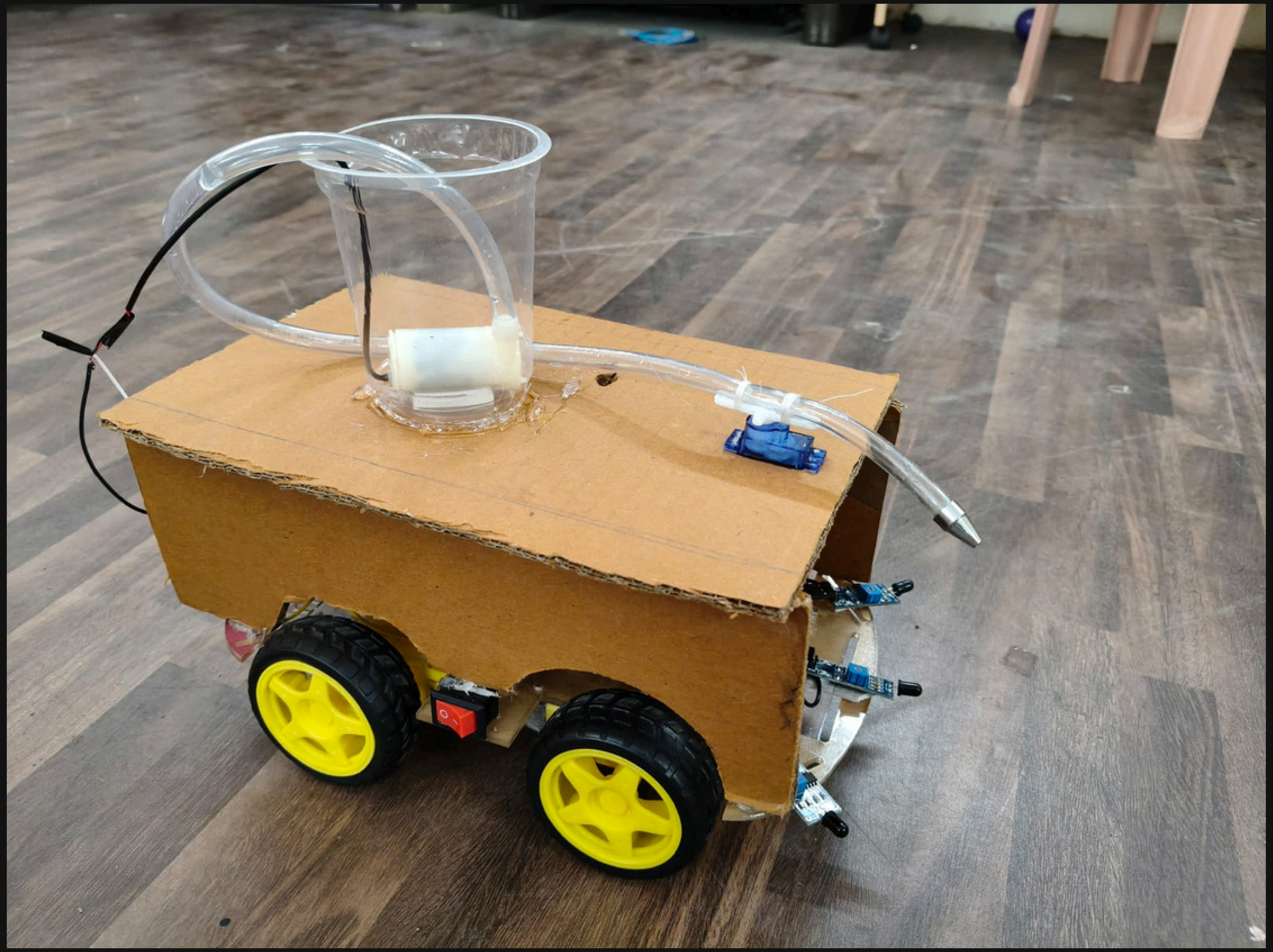
# 7. ALGORITHM

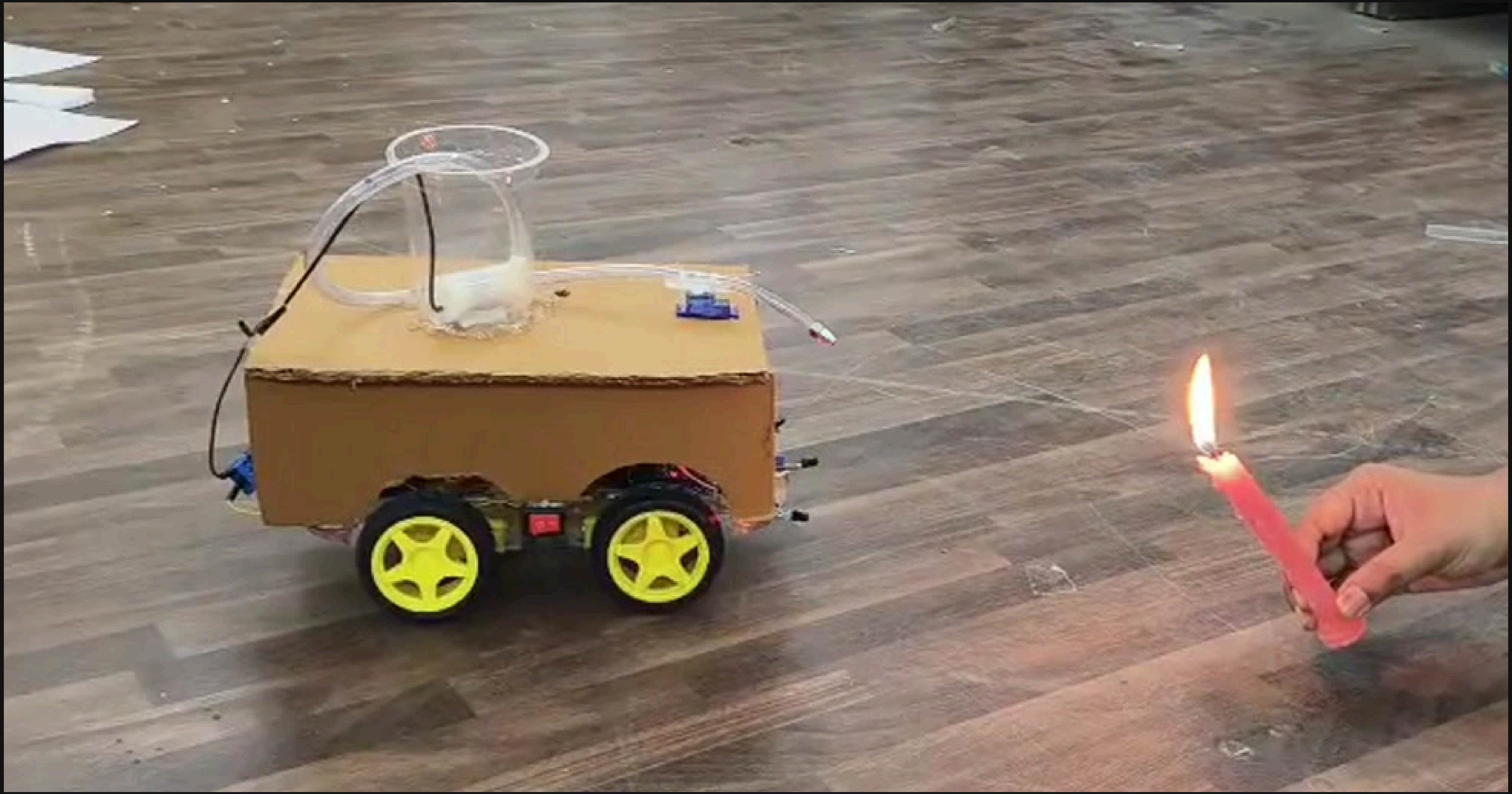


Start
Scan for fire
Sweep L → C → R → C
Fire found?
No
Move forward
400ms, then stop
Yes
Face the fire
Turn until center sensor active
Extinguish fire
Sweep servo 0°→180° while
fire_detected() == true
repeat

# 8. RESULT







# 9. APPLICATIONS

- Industrial Safety
- Warehouse Protection
- Home & Office Safety
- Forest Fire Monitoring
- Laboratory Safety
- Robotics & Research

# LIMITATIONS

- Limited Detection Range
- Slow Response in Large Areas
- False Alarm Issues
- Not Suitable for all Fire types

# 10. FUTURE SCOPE

- Multi-specturm sensor
- Faster microcontroller
- Human Detection & Rescue
- Improved Fire Suppression

# THANK YOU